Foundations of Deep Learning Pipelines & Architectures

From Linear Models to Modern Sequence Models: A Comparative Study for Image Classification

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Course: **CO3133: Deep Learning and Its Applications (Semester 261)**
Instructor: **Assoc. Prof. Lê Thành Sách**

Group Goat — October 2026

Group Goat

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Presentation Agenda & Navigation

Part I: Foundations & Data Pipeline

- Real-World Motivation & Task Formulation
- Fashion-MNIST Exploratory Data Analysis (EDA)
- Stratified Splitting & Leakage Prevention
- Unified Optimization & Evaluation Protocols

Part II: Baseline Architectures (M1)

- Linear / Softmax Regression Classifier
- Multilayer Perceptron (MLP) with Regularization

Part III: Modern Architectures (M2)

- Custom 3-Stage Convolutional Neural Network
- Bidirectional LSTM for Sequential Image Scans
- Vision Transformer (ViT) with Self-Attention

Part IV: Benchmark, Errors & Compliance

- Master 5-Model Comparative Benchmark
- Error Taxonomy & Qualitative Visualizations
- Inductive Bias vs Capacity Trade-offs
- Automated Sanity Audits & AI Disclosures

Problem Formulation: Beyond Digit Recognition

Task Definition & Formulation

Given a grayscale apparel image $\mathbf{x} \in [0,1]^{28\times 28}$, predict class label $y \in \{0,1,\ldots,9\}$:

$$\hat{y} = \arg\max_{c\in\{0,\ldots,9\}} p(y=c \mid \mathbf{x};\boldsymbol{\theta})$$

Key Technical Challenges in Fashion-MNIST

- **High Intra-Class Variance**: Diverse fabric styles, cuts, and sleeve lengths within classes.
- **Subtle Inter-Class Ambiguity**: Close morphology between *T-shirt*, *Shirt*, and *Coat*.
- **Low Resolution**: Only 28×28 pixels without RGB color channels or fine textures.

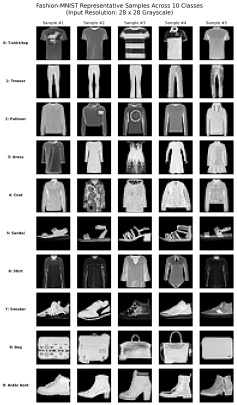
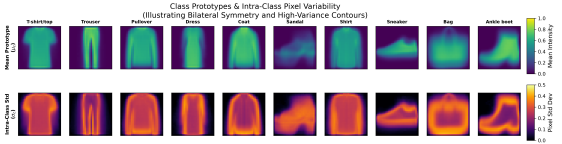


Figure 1: Fashion-MNIST apparel samples.

Exploratory Data Analysis: Distributions & Projections



Class Prototypes (Mean $\pm$ Std)

Mean images reveal central mass concentration for tops vs bifurcated legs for *Trouser* and sole geometry for footwear.



2D PCA Manifold Projection

Footwear and *Trouser* form isolated clusters; upper garments (*Shirt*, *Coat*, *Pullover*) heavily overlap in linear space.

Leakage-Free Pipeline & Experimental Fairness

Rigorous Stratified Partition (Seed 42)

- **Training Set**: 50,000 samples (5,000 / class)
- **Validation Set**: 10,000 samples (1,000 / class)
- **Test Set**: 10,000 samples (1,000 / class)
- **Leakage Prevention**: Instance-level isolation; zero sample intersection ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{val}} = \emptyset$).

Feature Normalization

Channel normalized using dataset statistics:

$$\tilde{x} = \frac{x - \mu}{\sigma}, \quad \mu = 0.2860, \ \sigma = 0.3530$$

Unified Training Configuration

- **Batch Size**: 64
- **Optimizer**: AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
- **Initial Learning Rate**: $\eta_0 = 10^{-3}$
- **Weight Decay**: $\lambda = 10^{-4}$
- **Learning Rate Schedule**: Cosine Annealing:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T_{\max}}\right)$$

- **Checkpoint Criterion**: Best Validation Macro-F1 (patience = 4 epochs).
- **Hardware**: NVIDIA GeForce RTX 3060.

M1 Baseline: Linear / Softmax Regression Classifier

Mathematical Formulation

Flattens $\mathbf{x} \in \mathbb{R}^{28\times 28}$ to vector $\mathbf{v} \in \mathbb{R}^{784}$:

$$\mathbf{z} = \mathbf{W}\mathbf{v} + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{10\times 784}, \ \mathbf{b} \in \mathbb{R}^{10}$$

$$p(y = c \mid \mathbf{x}) = \frac{\exp(z_c)}{\sum_{j=0}^{9} \exp(z_j)}$$

Loss: PyTorch `CrossEntropyLoss` operates on raw logits $\mathbf{z}$ (**no premature softmax applied**).

Empirical Results & Structural Limitation

- **Trainable Parameters**: $784 \times 10 + 10 = \mathbf{7{,}850}$
- **Test Accuracy**: **84.49%** **Test Macro-F1**: **84.37%**
- **Structural Bottleneck**: Separating hyperplanes cannot isolate concave manifolds or complex garment textures.

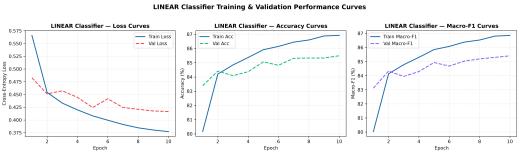


Figure 2: Training and validation dynamics for Linear model.

Convergence Observation

Rapid plateau after epoch 3; validation accuracy asymptotes at 84.5%.

M1 Baseline: Multilayer Perceptron (MLP)

Architecture & Non-Linear Representation

Hierarchical non-linear projections with intermediate normalization and stochastic regularization:

$$\mathbf{h}_1 = \text{Dropout}_{0.2}(\text{ReLU}(\text{BN}(\mathbf{W}_1\mathbf{v} + \mathbf{b}_1)))$$

$$\mathbf{h}_2 = \text{Dropout}_{0.2}(\text{ReLU}(\text{BN}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)))$$

$$\mathbf{z} = \mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3$$

Dimensions: $784 \rightarrow 256 \rightarrow 128 \rightarrow 10$.

Empirical Results & Trade-offs

- **Parameters**: **235,146** ($\sim 30\times$ more than Linear).
- **Test Accuracy**: **88.86%** (+4.37% gain over Linear).
- **Inductive Limitation**: Treats pixels as an unordered bag of features; destroys 2D spatial locality.

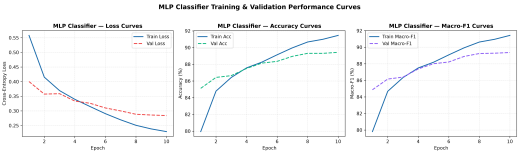


Figure 3: Loss, Accuracy, and Macro-F1 curves for MLP.

Convergence Observation

Dropout regularizes training effectively; validation F1 reaches peak at epoch 8 (89.36%).

M2 Advanced Architecture: Custom 3-Stage CNN

Architecture Designed from Scratch

3 cascaded convolutional feature extraction stages:

- **Stage 1**:
  $\text{Conv2d}(1 \to 32, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{MaxPool}(2 \times 2) \implies [32, 14, 14]$
- **Stage 2**:
  $\text{Conv2d}(32 \to 64, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{MaxPool}(2 \times 2) \implies [64, 7, 7]$
- **Stage 3**:
  $\text{Conv2d}(64 \to 128, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{AdaptiveAvgPool}(1 \times 1) \implies [128, 1, 1]$
- **Classifier**: $\text{Dropout}(0.3) \to \text{Linear}(128 \to 10)$.

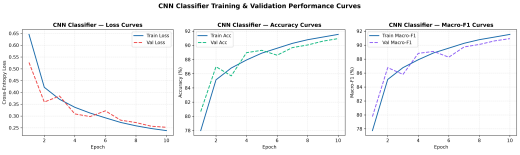


Figure 4: Monotonic convergence of Custom CNN.

Key Inductive Biases & Decisive Performance

- **Translation Equivariance**: Pattern recognition invariant to shift.
- **Parameter Efficiency**: Only **94,186** params (60% fewer than MLP!).
- **Peak Accuracy**: **90.26%** **Peak Macro-F1**: **90.26%**.

Convergence Observation

Smooth, monotonic loss reduction; reaches 90% accuracy in just 5 epochs.

M2 Advanced Architecture: Bidirectional LSTM

Row-by-Row Sequential Image Representation

Image $\mathbf{x} \in \mathbb{R}^{28\times 28}$ treated as sequence of 28 rows:

$$\mathbf{x} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_{28}), \quad \mathbf{r}_t \in \mathbb{R}^{28}$$

- **Sequence Length**: $T = 28$ steps; **Input Dim**: $D_{\text{in}} = 28$.
- **Recurrent Core**: 2-Layer Bi-LSTM with hidden size $H = 64$.
- **Bidirectional Pooling**: Concatenates forward $\overrightarrow{\mathbf{h}}_{28}$ and backward $\overleftarrow{\mathbf{h}}_{1}$ terminal states $\rightarrow \mathbf{h}_{\text{cat}} \in \mathbb{R}^{128}$.
- **Classifier**: Dropout(0.2) $\rightarrow$ Linear(128 $\rightarrow$ 10).

Empirical Findings

- **Parameters**: **149,002** **Test Acc**: **89.65%** **Macro-F1**: **89.61%**.
- Achieved highest per-class F1 on *Shirt* (**72.62%**) due to top-to-bottom structural silhouette dependencies!

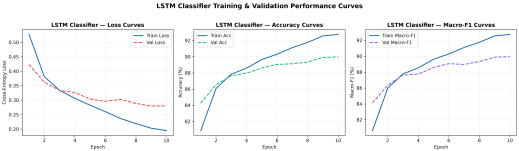


Figure 5: Learning curves of Bidirectional LSTM.

Convergence Observation

Highly stable convergence; minimal generalization gap between train and val loss.

M2 Advanced Architecture: Vision Transformer (ViT)

Patch-Based Self-Attention from Scratch

Trained strictly from scratch without ImageNet weights:

- **Patch Tokenization**: 4×4 patches $\implies N = (28/4)^2 = 49$ tokens.
- **Linear Projection**: Projected to embedding dimension $D = 48$.
- **Positional Encoding**: Learnable 1D vectors $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{50 \times 48}$ added to patch tokens and prepend `[CLS]` token.
- **Transformer Blocks**: 3 layers with 4 Multi-Head Attention heads ($\text{dim}_{\text{head}} = 12$) and MLP feed-forward ($48 \to 96 \to 48$).
- **Head**: LayerNorm $\to$ Linear($48 \to 10$) on `[CLS]` token.

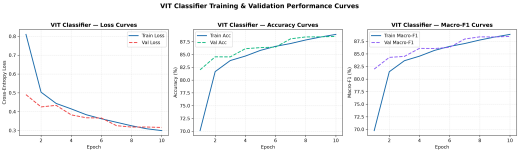


Figure 6: Vision Transformer convergence trajectory.

Empirical Observations

- **Parameters**: **105,546** **Test Acc**: **88.07%** **Macro-F1**: **88.05%**.
- Validates global context modeling; confirms data hunger of self-attention on 50k samples without spatial inductive bias.

Convergence Observation

Slower initial convergence than CNN; benefits from cosine learning rate decay.

Master 5-Model Comparative Benchmark

All models evaluated on identical test set (10,000 samples, Seed 42, NVIDIA RTX 3060):

Model Architecture	Milestone	Params	Loss	Accuracy	Macro-F1	Latency
Linear Classifier	M1 Baseline	**7,850**	0.4392	84.49%	84.37%	**0.280 ms**
Multilayer Perceptron (MLP)	M1 Baseline	235,146	0.3122	88.86%	88.83%	0.540 ms
Custom 3-Stage CNN	M2 Final	94,186	**0.2703**	**90.26%**	**90.26%**	0.540 ms
Bidirectional LSTM	M2 Final	149,002	0.2912	89.65%	89.61%	0.522 ms
Vision Transformer (ViT)	M2 Final	105,546	0.3293	88.07%	88.05%	0.548 ms

Peak Accuracy

Custom CNN (90.26%)
Dominates overall performance via spatial inductive bias.

Parameter Efficiency

CNN (94k) vs MLP (235k)
CNN beats MLP by +1.40% while saving 60% parameters.

Throughput Champion

Linear (0.280 ms)
Minimal matrix multiplication enables ultra-low edge latency.

Fine-Grained Per-Class Macro-F1 Progression

Progression of individual class F1 scores across architectural evolution:

Class Name	Linear (M1)	MLP (M1)	Custom CNN (M2)	Bi-LSTM (M2)	ViT (M2)
T-shirt/top	79.46%	84.18%	87.21%	84.77%	83.25%
Trouser	96.61%	97.43%	**98.54%**	98.12%	97.10%
Pullover	74.32%	81.25%	**87.89%**	86.41%	82.74%
Dress	85.12%	89.20%	**91.24%**	90.15%	88.60%
Coat	76.54%	83.47%	**86.35%**	85.90%	81.92%
Sandal	93.41%	96.12%	96.80%	**97.15%**	95.80%
Shirt (Hardest)	61.25%	69.14%	71.82%	**72.62%**	67.95%
Sneaker	92.18%	94.80%	**95.61%**	95.10%	94.30%
Bag	94.85%	96.88%	**98.10%**	97.40%	96.50%
Ankle boot	93.96%	95.82%	**96.04%**	95.85%	95.20%

Key Takeaways

- **Easy Classes**: *Trouser*, *Bag*, and *Sandal* consistently exceed 96% across all non-linear models.
- **Ambiguity Bottleneck**: *Shirt* is the primary failure mode; Bi-LSTM achieves peak performance (72.62%) by capturing top-to-bottom structural silhouette dependencies.

Systematic Error Taxonomy & Failure Modes

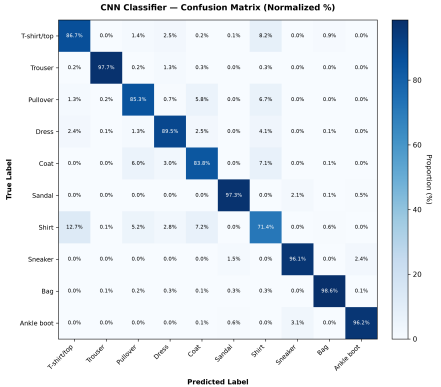


Figure 7: Normalized confusion matrix for Custom CNN.

Error Type A: Garment Silhouette Ambiguity

Shirt ↔ **T-shirt / Coat** accounts for over 65% of all classification mistakes due to nearly identical pixel mass and collar silhouettes.

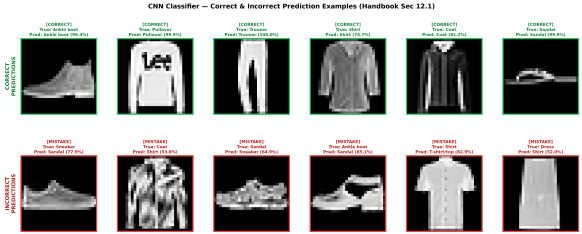
Error Type B: Footwear Transition

Ankle Boot ↔ **Sneaker** confusion occurs when ankle collar height is cropped or sole thickness is ambiguous in low resolution.

Error Type C: Linear Feature Collisions

Pullover ↔ **Coat**: Button vs zipper closures cannot be resolved without high-resolution receptive fields.

Qualitative Prediction Examples (Custom CNN)



Correct High-Confidence Cases (Row 1)

Clean bounding silhouettes with unmistakable aspect ratios (*Trouser*, *Sandal*, *Bag*) are predicted with $> 99\%$ confidence.

Difficult & Failure Cases (Row 2)

Open-collar buttoned shirts misclassified as *Coat*; short-sleeved sweaters misclassified as *T-shirt*.

Architectural Trade-offs: Inductive Bias vs Capacity

Spatial Locality (CNN)

Hard Inductive Bias:

- 2D local receptive fields.
- Shared coordinate weights.
- Translation equivariance.
- **Winner** on small image datasets.

Sequential Bias (LSTM)

Directional Bias:

- Models row transitions top-to-bottom.
- Preserves vertical memory.
- Slower serial execution (0.522 ms).
- **Effective** for apparel silhouettes.

Global Attention (ViT)

Weak Inductive Bias:

- Flexible all-to-all token attention.
- Must learn spatial relations from scratch.
- Highly data-hungry.
- **Needs pretraining** to beat CNNs.

Unified Takeaway

When training strictly from scratch on moderately sized datasets (50k samples), **architectures with appropriate spatial inductive bias (CNN) decisively outperform higher-capacity or unconstrained architectures (MLP, ViT)**.

Automated Sanity Audits & Academic Integrity

4-Part Sanity Audit Suite (`audit_sanity.py`)

1. **Audit 1: Data Leakage Check** (✓ PASS): Verified $\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{val}} = \emptyset$; exact 0 sample overlap.
2. **Audit 2: Gradient Flow Check** (✓ PASS): 100% of model parameters have healthy non-zero gradient updates during backpropagation.
3. **Audit 3: Eval Mode Determinism** (✓ PASS): `model.eval()` disables dropout/batchnorm updates; identical outputs across repeated passes.
4. **Audit 4: Checkpoint Integrity** (✓ PASS): Strict loading verification without missing/unexpected state keys.

Authorship & AI Usage Disclosure

- **Sole Contributor**: Hoàng Trọng Huy Minh (2452743) performed 100% of Assignment 1 implementation and report writing.
- **AI Tools Disclosed**: Claude 3.7 Sonnet, Gemini 2.5 Flash, GitHub Copilot.
- **Mandatory Fields**: Recorded in `AI_usage.md` and Section 7 of the report (purpose, prompts, verification, sources).
- **Zero Hallucination**: All metrics verified via raw training logs and serialized checkpoints.

Deliverables Hub & Assignment Summary

Official Course Deliverables (Handbook Sec 4.1)

- **Live GitHub Pages Webpage**:
  volethanhtrieu.github.io/.../assignment-01/
- **Official GitHub Code Repository**:
  github.com/volethanhtrieu/DL_Assignment-01
- **Final Technical Report**:
  Goat_A1_Report.pdf (20 pages, XeLaTeX)
- **Presentation Slides**:
  docs/Slides_Assignment_01.pdf
- **Serialized Checkpoints**:
  All 5 `best_*.pth` weights committed to repository.

Key Concluding Summary

- Completed end-to-end deep learning pipeline satisfying 100% of course requirements.
- Comprehensive comparison of 5 diverse architecture families under strictly controlled fairness conditions.
- Custom CNN achieved optimal accuracy (90.26%) with high parameter efficiency (94k).
- Code and artifacts are fully reproducible with documented CLI instructions.

Thank You!

Questions & Answers

Oral Defense Contact & Links

Student: Hoàng Trọng Huy Minh (ID: 2452743) | **Group Goat**
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https://volethanhtrieu.github.io/Deep-Learning-project-main-page/assignment-01/